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Fusion of Dermoscopic Images and Patient Metadata for Skin Lesion Classification: A Sample-Level Benefit-Cost Trade-off Analysis

1. Problem Formulation & Research Question

Skin lesion screening is critical for early melanoma diagnosis. Clinicians combine dermoscopic visuals with patient metadata (age, sex, lesion location). Prior multimodal studies fuse images and metadata for 7-class HAM10000 classification.

Existing work relies heavily on global metrics (accuracy, macro-F1, AUC), even with rigorous lesion-id splitting. Aggregate metrics can mask heterogeneous model behaviour across patient subgroups. Global performance gains may come at the cost of worse predictions for certain populations, raising clinical safety and fairness concerns.

Research Question: Under lesion-level train-val-test split on HAM10000, is performance change from metadata fusion evenly distributed across test samples? Do some subgroups benefit while others degrade?

Hypothesis: Metadata-driven performance shifts are non-uniform. Even improved global metrics may cluster benefit and harm within specific disease, age or location subgroups; global-only evaluation is insufficient for medical-AI assessment.

2. Prior Work & Background

Released 2018, HAM10000 includes dermoscopic images and patient metadata. Early studies (2018-2020) used only images. Starting in 2021, multimodal models fused images with demographic features. Many early papers split by raw image, producing leakage-inflated results.

Recent work adopts lesion-id stratified splits and global-oriented ablation. However, nearly all evaluations focus on aggregate performance. Few decompose test samples into benefited / neutral / harmed groups or examine metadata-driven subgroup risks. This project fills this gap via sample-wise trade-off analysis.

3. Dataset Description

Public HAM10000 (Kaggle, no IRB): 10015 dermoscopic images, seven diagnostic classes.

Modality1: lesion dermoscopic images

Modality2: tabular metadata: age, sex, localization, lesion_id, diagnosis label dx

Preprocessing: image resize, normalization, random flip-rotation augmentation. Age standardized; sex/localization one-hot encoded. We use stratified lesion-id-level split: all images of one lesion stay within one partition.

4. Planned Methods & Evaluation Strategy

Supervised 7-class classification with two models trained on identical train/validation partitions.

Baseline (image-only): Pre-trained ResNet-50 with replaced classification head,

trained using cross-entropy loss.

Multimodal model: Two-stream network. ResNet-50 extracts visual features; metadata is projected to matching dimension via a small MLP. Mid-level feature concatenation followed by fully-connected and Dropout layers for end-to-end training.

Ablations: (1) image-only; (2) image+all metadata; (3) image+age only; (4) image+localization only.

Core sample-level & subgroup analysis: On held-out test set, compare prediction correctness between baseline and multimodal model to generate three disjoint subsets: Benefit (unimodal wrong, multimodal correct), Harm (unimodal correct, multimodal wrong), Neutral (both correct / both wrong). We compare disease, age, sex and lesion-location distributions across subsets and compute standard metrics within demographic subgroups.

Metrics: Accuracy, macro-F1, one-vs-rest AUC, confusion matrix. No new metrics are created; novelty comes from stratified evaluation. This work aligns with course themes: medical imaging, multimodal integration, fairness and failure-mode analysis.

5. Milestone Timeline & Group Responsibilities

Week	Milestone	Member
1	Dataset download, cleaning, EDA, lesion-id split; literature review; finalize proposal	A, D
2	Build unimodal & multimodal models; tune hyper-parameters; export test-set predictions	B
3	Implement Benefit-Neutral-Harm partitioning; subgroup statistics & plots; run ablations; apply fallback if required	C
4	Write final report; assemble figures; prepare 10-min presentation; group rehearsal; individual reflections	All, led by D

Group responsibilities summary:

A: dataset download, cleaning, preprocessing, lesion-id split, EDA.

B: develop unimodal baseline and multimodal fusion model, training, tuning, export test-set predictions

C: implement sample-level partition logic, subgroup statistical analysis, visualization

D: literature review, proposal & final report writing, presentation slide preparation, GitHub repository maintenance

6. Expected Challenges

Training instability: the strong pre-trained image branch may dominate joint training, limiting metadata learning. Late-fusion fallback is prepared.

Uncertain experimental outcomes: multimodal performance may improve, stay similar or degrade relative to baseline. Our sample-level trade-off analysis remains valid regardless of global-level outcome.

Subset size noise: Benefit / Harm subsets could contain limited samples; we will interpret observations as correlational patterns instead of strong causal conclusions and discuss this limitation explicitly.

References

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